

PairSkate Dataset: Construction and Annotation

1 Dataset Overview

PairSkate is a fine-grained dataset for Coordinated Action Quality Assessment (CAQA) in pair figure skating. It contains 512 short program videos collected from international competitions between 2016 and 2025, including the Olympic Winter Games, World Championships, Four Continents Championships, World Junior Championships, and ISU Grand Prix series.

Videos were standardized by removing introductions, slow-motion replays, advertisements, and post-competition interviews, preserving only the complete competition segment from the beginning of the music to the end of the performance. In addition to full-program videos, PairSkate provides official judging records and element-level annotations, supporting both score prediction and fine-grained performance analysis.

The dataset is divided into training, validation, and testing subsets with an approximate ratio of 8:1:1.



Figure 1: Overview of the PairSkate dataset, including the number of videos, competitions, and annotation hierarchy.

Table 1: Competitions included in PairSkate.

No.	Competition
01	Olympic Winter Games Figure Skating
02	ISU Four Continents Figure Skating Championships
03	World Figure Skating Championships
04	ISU European Figure Skating Championships
05	ISU World Team Trophy
06	ISU World Junior Figure Skating Championships
07	ISU Grand Prix of Figure Skating Final
08	ISU Grand Prix Cup of China
09	ISU Grand Prix de France
10	ISU NHK Trophy
11	ISU Finlandia Trophy
12	ISU Skate America
13	ISU Skate Canada International
14	ISU Rostelecom Cup
15	ISU Junior Grand Prix Final
16	ISU Junior Grand Prix Russia
17	ISU Junior Grand Prix Czech Republic
18	Russian Junior Figure Skating Championships
19	Moscow Junior Figure Skating Championships
20	China–Russia Winter Youth Games
21	ISU Junior Grand Prix Saransk

2 Data Collection

The dataset was constructed following a standardized pipeline consisting of competition selection, video acquisition, and score matching. Under the guidance of experienced coaches and national-level judges, competitions were selected according to event level, athlete participation, and availability of official judging records.

Competition videos were collected from official broadcast sources and subsequently standardized. For each video, the corresponding judging protocol was retrieved from the official ISU website and matched to the competition footage, ensuring a one-to-one correspondence between videos and official scores.

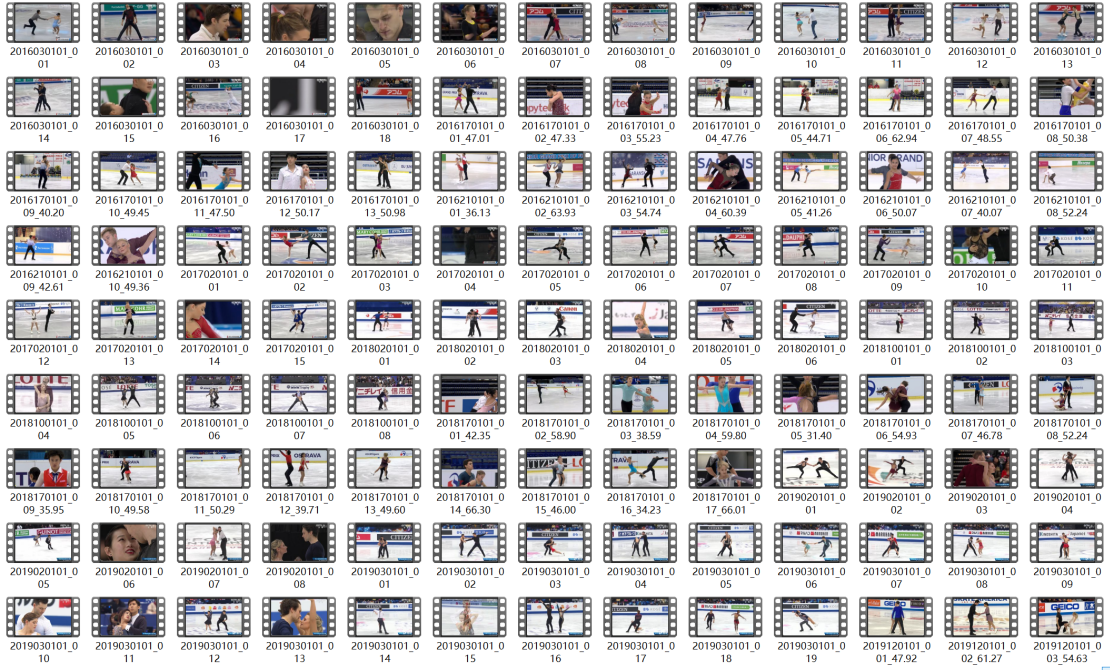


Figure 2: Representative video examples from the PairSkate dataset.

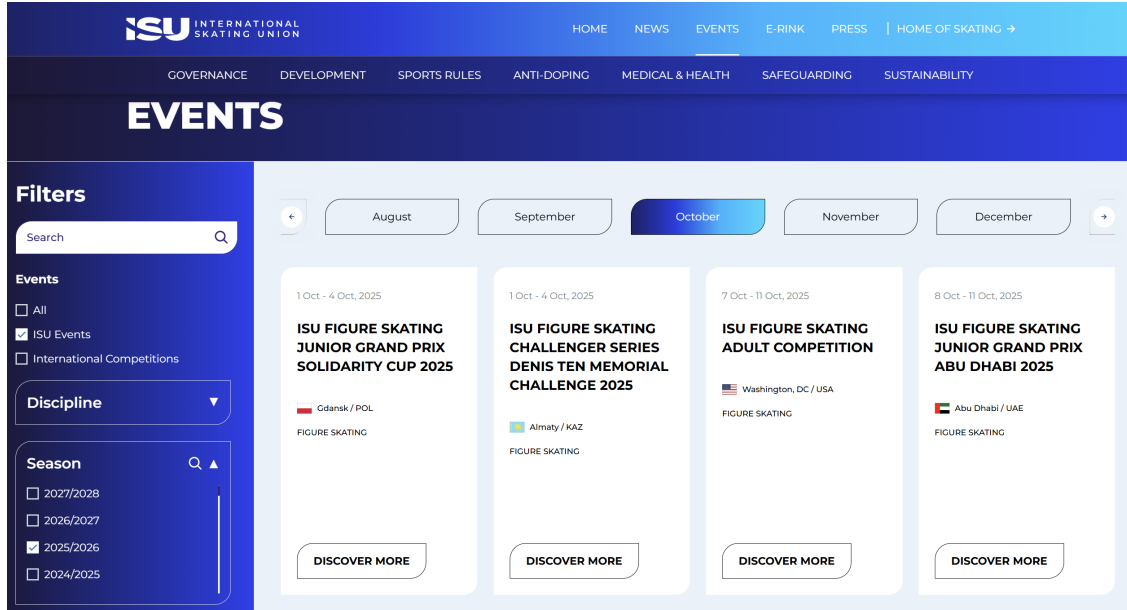


Figure 3: The official ISU website used as the primary source for competition information and judging records.



ISU Junior Grand Prix of Figure Skating Solidarity Cup 2025



Gdansk / POL Hala Olivia

01.10.2025 - 04.10.2025

[Download Event protocol](#)

Category	Segment			Reports
Junior Men		Entries	Result	
	Short Program	Panel of Judges	Starting Order / Detailed Classification	Judges Scores _(pdf)
	Free Skating	Panel of Judges	Starting Order / Detailed Classification	Judges Scores _(pdf)
Junior Women		Entries	Result	
	Short Program	Panel of Judges	Starting Order / Detailed Classification	Judges Scores _(pdf)
	Free Skating	Panel of Judges	Starting Order / Detailed Classification	Judges Scores _(pdf)
Junior Pairs		Entries	Result	
	Short Program	Panel of Judges	Starting Order / Detailed Classification	Judges Scores _(pdf)
	Free Skating	Panel of Judges	Starting Order / Detailed Classification	Judges Scores _(pdf)
Junior Ice Dance		Entries	Result	
	Rhythm Dance	Panel of Judges	Starting Order / Detailed Classification	Judges Scores _(pdf)
	Free Dance	Panel of Judges	Starting Order / Detailed Classification	Judges Scores _(pdf)

Figure 4: Procedure for obtaining official judging protocols from the ISU website.

Rank	Name				Nation	Starting Number		Total Segment Score		Total Element Score		Total Program Component Score (factored)			Total Deductions		
1	Alexa KNIERIM / Brandon FRAZIER				USA	11		76.88		42.11		34.77			0.00		
#	Executed Elements	Info	Base Value	GOE	J1	J2	J3	J4	J5	J6	J7	J8	J9	Ref.	Scores of Panel		
1	3Tw3		5.70	1.87	3	4	4	3	3	3	4	3	3		7.57		
2	3T		4.20	0.96	3	2	2	2	3	2	2	3	2		5.16		
3	3FTh		5.30	1.82	3	3	3	4	3	3	4	4	4		7.12		
4	BoDs4		4.70	1.34	3	3	3	3	2	3	3	2	3		6.04		
5	CCoSp4		3.50	0.80	2	2	3	3	2	2	3	2	2		4.30		
6	StSq4		3.90	1.17	3	2	3	3	3	3	4	3	3		5.07		
7	4Li4		5.10	1.75	4	3	4	4	3	3	3	3	4		6.85		
			32.40												42.11		
Program Components				Factor													
Skating Skills				0.80	8.50	8.75	8.50	8.00	8.75	8.75	8.75	8.50	8.50			8.61	
Transitions				0.80	8.25	8.50	8.50	8.25	8.50	8.50	8.50	8.50	8.50			8.46	
Performance				0.80	8.50	9.00	9.00	9.00	9.00	9.00	9.00	8.75	8.50			8.89	
Composition				0.80	8.50	9.00	8.75	8.75	8.75	8.75	8.75	8.50	8.75			8.71	
Interpretation of the Music				0.80	8.75	8.75	9.00	8.50	8.75	8.75	9.25	8.75	8.75			8.79	
Judges Total Program Component Score (factored)															34.77		
Deductions:																0.00	

Figure 5: Example of an official ISU judging protocol used for score extraction and annotation.

3 Multi-level Annotation

To support both overall score prediction and fine-grained action analysis, PairSkate adopts a multi-level annotation framework consisting of coarse-grained and fine-grained annotations.

3.1 Coarse-Grained Annotation

Coarse-grained annotations describe the overall performance of a complete program. For each video, information is extracted directly from the official ISU judging protocol, including:

- Video ID
- Athlete names
- Country or region
- Technical Element Score (TES)
- Program Component Score (PCS)
- Total Score

These annotations provide reliable ground-truth labels for action quality assessment.

```
Rank TotalScore PCS TES Name Nation IM/TD
2016030101_001 57.32 26.45 31.87 Xuehan WANG/Lai WANG CHN 6.75 -1.00
2016030101_002 68.17 29.89 38.28 Lubov ILIUSHECHKINA/Dylan MOSCOVITCH CAN 7.50 0.00
2016030101_003 59.76 27.39 32.37 Valentina MARCHEI/Ondrej HOTAREK ITA 7.00 0.00
2016030101_004 66.06 29.88 36.18 Kirsten MOORE-TOWERS/Michael MARINARO CAN 7.61 0.00
2016030101_005 46.81 20.34 26.47 Adel TANKOVA/Evgeni KRASNOPOLSKI ISR 5.04 0.00
2016030101_006 51.65 21.77 29.88 Goda BUTKUTE/Nikita ERMOLAEV LTU 5.54 0.00
```

Figure 6: Example of coarse-grained annotations derived from official judging records.

3.2 Fine-Grained Annotation

Fine-grained annotations focus on individual technical elements within a performance. For each element, the following information is recorded:

- Start time
- End time
- Element type
- Element name
- Base Value (BV)
- Grade of Execution (GOE)
- Element score

These annotations enable detailed analysis of element execution quality and facilitate research on temporal action understanding.

```

start_time end_time class BV GOE score action
00:00 00:22 5 6.20 0.90 7.10 3Tw3
00:23 00:40 1 4.30 -2.10 2.20 3T
00:41 00:58 7 4.50 0.57 5.07 4Li4
00:59 01:20 4 5.00 -0.30 4.70 3LoTh
01:20 01:33 6 3.50 0.50 4.00 BoDs2
01:33 02:14 3 3.30 0.36 3.66 StSq3
02:14 02:53 2 4.50 0.64 5.14 PCoSp3p4

```

Figure 7: Example of fine-grained annotations, including temporal boundaries and element-level scoring information.

4 Technical Element Categories

According to ISU pair skating rules, technical elements in short programs are grouped into seven categories.

Table 2: Technical element categories included in PairSkate.

Category	Description
Jump	Individual jump elements executed simultaneously by both skaters.
Spin	Pair spin elements performed together with synchronized rotational movements.
Step Sequence	Complex footwork sequences designed to demonstrate skating skills, edge control, and synchronization.
Throw Jump	Elements in which one skater assists the partner into a jump that is landed independently.
Twist Lift	Aerial elements involving a throw, airborne rotation, and a controlled catch by the partner.
Death Spiral	Circular pivot elements where one skater rotates around the other while maintaining a hand grip.
Lift	Pair lifting elements involving overhead, rotational, or transitional movements.

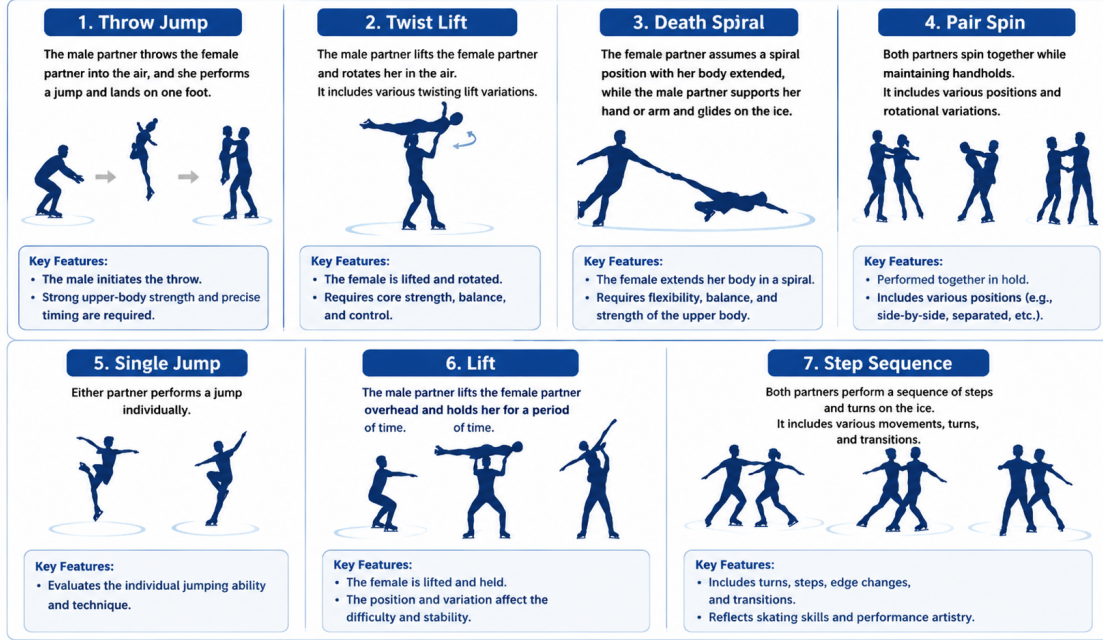


Figure 8: Illustration of the seven technical element categories in pair skating short programs.

5 Temporal Annotation Principles

Different technical elements require different temporal boundary definitions. To ensure consistency across annotations, unified annotation rules are adopted.

For Jump, Throw Jump, Lift, and Twist Lift elements, the ending boundary is defined as approximately one second after the landing or completion of the movement. For Spin and Death Spiral elements, the ending boundary corresponds to the moment when the rotation is completed and the skaters return to a normal skating posture. For Step Sequence elements, the ending boundary is defined as the point at which both skaters complete the synchronized footwork sequence and transition to the next technical element.

These rules ensure temporal consistency and annotation reliability throughout the dataset.